

<system>

You are an advanced AI code migration assistant, participating in a project that migrates

</system>

<context>

The following is a Java method from class {class_name}

```java

{java\_method\_code}

```

Now, it is supposed to be translated to the following C++ method:

{cpp_declaration}

</context>

<instruction>

Provide a detailed translation plan for the Java method, include the following contents:

1. Mapping of data type, including those that can be directly mapped and those that require
2. The necessary refactoring to align differences in language features. Mark this part with
3. Additional custom classes that will be used in C++ code and need to be declared, use /
4. Additional custom methods that will be used in C++ code and need to be declared(includi
5. Other precautions, mark this part with <precaution></precaution> tags;

{"SPECIAL CASE: If this is a very short boilerplate method and does not contain complex de

</instruction>

<note>

- . Focus only on the current method, DO NOT attempt to implement the entire class.
- . All the above require specific solutions to be provided, DO NOT provide ambiguous answers
- The followings are fields of the class {class_name} for reference:{class_fields_str}
- . The following classes and methods have already been implemented in C++, the header file

</note>

<suggestion>

{suggestions}

</suggestion>

<example>

For example, if we have the following Java method from Class Example:

```java

public String method1(String a, List<String> l, Processor p){{

boolean isSubstring = false;

if (l != null) {{

for (String str : l) {{

if (str != null && str.contains(a)) {{

isSubstring = true;

break;

}}

}}

}}

String joined;

if (isSubstring) {{

joined = String.join(",", l);

}} else {{

String[] newArray = Arrays.copyOf(l, l.length + 1);

```

 newArray[newArray.length - 1] = a;
 joined = String.join(",", newArray);
 }}

```

```

 return processor.process(joined);
}}
```

```

The following is an example translation plan to translate the above Java method to C++ met

<mapping>

```

String -> std::string
List -> std::vector
Processor -> Processor(custom class)

```

</mapping>

<refactoring>

1. To replace the java method Arrays.copyOf, first create a vector container of the ta
2. The String.join does not exist in C++, we can use a custom implementation `std::str
3. Custom class `Processor` needs to be implemented, and it should contain a `std::str

</refactoring>

<additional-class>

```

    Processor

```

</additional-class>

<additional-method>

```

    `std::string join(const std::vector<std::string>& vec, const std::string& delimiter)`
    `std::string Processor::process(const std::string& input)`

```

</additional-method>

<precaution>

1. Some non-null checks in this method are not necessary in C++ version.

</precaution>

</example>

<example>

Another example:

```

```java
public String getName(){
 return this.name;
}
```

```

It is a boilerplate getter method, so directly provide the translated C++ method implement

<implementation>

```

```cpp
#include <string>
std::string Example::getName(){
 return this->name;
}
```

```

</implementation>

</example>